

RITWIK UPADHYAY

Bhopal, Madhya Pradesh

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LinkedIn

GitHub

Portfolio

Summary

Full-stack Developer and B.Tech CSE student at LNCT Bhopal skilled in Java, Python, JavaScript, React.js, Node.js, Spring Boot, AI/ML, LangChain, TensorFlow, REST APIs, DSA, DBMS, and Operating Systems, with national-level hackathon recognition.

Education

Lakshmi Narain College of Technology (LNCT)

B.Tech in Computer Science Engineering CGPA: 8.48/10

Sep 2023 – May 2027

Bhopal, Madhya Pradesh

Indian Institute of Technology Madras (IIT Madras)

B.S. in Data Science and Applications (Foundation Level) CGPA: 8.88/10

May 2023 – Mar 2025

Chennai, Tamil Nadu

Class XII (ISC)

Percentage: 96.75%

2022

Lucknow, Uttar Pradesh

Class X (ICSE)

Percentage: 96.80%

2020

Lucknow, Uttar Pradesh

Skills

Languages: Java, Python, JavaScript

Web Development: HTML5, CSS3, Node.js, Express.js, React.js, Spring Boot, REST APIs

Databases: MongoDB, SQL (MySQL), MongoDB Atlas

AI/ML: TensorFlow, Transfer Learning (MobileNetV2), RAG, LangChain, Vector Databases, Embeddings

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Tools & Platforms: Git, GitHub, VS Code, Render, Postman

Soft Skills: Problem Solving, Team Collaboration, Leadership, Effective Communication, Attention to Detail

Projects

Noted – Full-Stack Note Application | Node.js, Express.js, MongoDB, Passport.js, REST API | Live | GitHub **Dec 2025**

- Built a **full-stack CRUD web app** with **Google OAuth 2.0**, REST APIs, session management, and real-time note search using MongoDB Atlas.
- Deployed on Render with secure environment variables and **zero-downtime** cloud-hosted MongoDB Atlas persistence.

AlgoMentor – RAG-Based DSA Tutor | Python, LangChain, Vector Database, LLMs, Embeddings | GitHub **Feb 2026**

- Built a **RAG pipeline** using embeddings, vector search, and LLMs for context-aware DSA solutions with ~40% fewer irrelevant responses.
- Integrated **LangChain orchestration** and semantic retrieval across a curated DSA knowledge base.

PlantCare AI – Crop Disease Detection System | Python, TensorFlow, MobileNetV2, Flask, Transfer Learning **Mar 2026**

- Developed an **ML-powered web app** using **MobileNetV2 transfer learning** for multi-class plant disease classification.
- Implemented image preprocessing, model inference, and real-time prediction via a **Flask REST API**.

StarBlaze – Browser-Based Retro Space Shooter | HTML5, CSS3, JavaScript, Canvas API | Live | GitHub **Nov 2025**

- Led a team of 4** to build a browser-based arcade game with collision detection, dynamic scoring, and smooth 60fps Canvas API animations.
- Designed a GameBoy-inspired UI with custom pixel art and **Git-based collaborative development**.

Achievements

AB PM-JAY Auto-Adjudication Hackathon | NHA, IndiaAI Mission & IISc Bengaluru

May 2026

- National-level hackathon by NHA, IndiaAI Mission & IISc Bengaluru; built an AI solution for **Radiological Image-Based Condition Detection & Report Correlation** (PS-2) as a team of 3, scoring **0.48** on the evaluation metric.

Yuva Sankalp Ideathon 2026 – 2nd Position | University of Delhi, NSS – Keshav Mahavidyalaya

April 2026

- Secured **2nd Position** among all competing teams in a national-level ideathon; presented an innovative, socially impactful solution before a panel of industry and academic judges.

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Oracle

Nov 2025

Score: 98%

Data Structures and Algorithms using Java

Infosys Springboard

Nov 2025